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An Ante Hoc Enhancement Method for Image-Based Complex Financial Table Extraction

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Featured Application: Image-based complex table extraction.

Abstract: In the field of finance, the table is a common form of data organization. Extracting data from these tables in large quantities is a fundamental and important task for researchers. However, this can be a challenging task, as many tables exist in unstructured forms, such as scanned images in PDFs, rather than forms which can be easily processed, such as Excel spreadsheets. In recent years, a large number of table extraction methods utilizing heuristic algorithms or deep learning models have been proposed to free people from manual processing tasks, which are time-consuming and troublesome. Although existing methods achieve high levels of accuracy in processing some kinds of tables, they often fail to achieve optimal results when extracting complex financial tables with multi-line text and missing demarcation lines. In this article, we propose an enhancement method for image-based complex table extraction. This method consists of two modules: a split module and a filter module. The split module uses an OCR (optical character recognition) model to locate text regions, and a heuristic algorithm to obtain candidate demarcation lines. The filter module is based on a text semantic matching model and another heuristic algorithm. The experimental results show that the use of the proposed method can significantly improve the performance of different table extraction methods, with increases in F1 scores of between 5.10 and 14.36 points being recorded.

Keywords: table extraction; information extraction; text matching; artificial intelligence

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1. Introduction

Tables are widely used for representing and arranging data due to their high visibility and ease of use. In the field of finance, tables are commonly found in financial statements, audit reports, and annual reports. Financial professionals rely on these tables for industry research reports, credit rating reports, and competitor analysis reports. Academic researchers conducting financial research also rely heavily on tables as crucial sources of information.

However, in many cases, tables do not exist in structured forms (such as Word tables, Excel spreadsheets, or relational database tables), but in unstructured forms (such as scanned images in PDFs) that do not contain the structural information of the tables. Although the use of unstructured forms for tables does not affect people's ability to read and understand them, it does pose significant challenges for information extraction, as the mining and application of data from such tables is hindered.

Traditionally, extracting tables from images has been a manual process. However, this method is inefficient, costly, and prone to errors. Therefore, the problem has arisen of how best to convert tables in unstructured forms, such as images, into structured forms, such as Excel spreadsheets or HTML sequences. This process is known as table extraction (TE), and involves three steps: table detection, table structure recognition, and table recovery. Table detection is the process of identifying and locating tables in an image. Table structure recognition involves identifying the structure of a table, which is the most important and challenging aspect of table extraction. Table recovery involves recognizing text and inserting it into the cells of the table. In recent years, extensive research has been conducted on table extraction, significantly improving the performance of table extraction models.

In terms of table extraction, tables are classified into two categories: simple tables and complex tables. Simple tables are those with complete demarcation lines or those without spanning cells or cells with wrapped text (text that is split into several text lines). Complex tables are those with missing demarcation lines, spanning cells, or cells with wrapped text. The first techniques developed for table extraction were heuristic methods or simple machine learning methods [1–7]. These methods may yield good results in extracting simple tables, but they perform poorly when dealing with complex tables. In recent years, methods based on deep learning models [8–19] such as RCNN (region-based CNN) and FCNs (fully convolutional networks) have made significant progress in table extraction. However, their performance in extracting complex tables with wrapped text is still inadequate for practical use in specialized fields. For instance, in the table shown in Figure 1, the first cell in the second row has three text lines, but the line spaces within this cell are the same as those between cells, causing it to easily be recognized as three cells. To enhance the extraction of complex financial tables, it is essential to consider the semantic relationships between neighboring text lines and to provide the semantic information as an instruction for extraction methods to help them recognize those cells with wrapped text. Existing methods often fail to achieve optimal results because they either solely consider visual features or use a representation-based model to encode text and mix textual semantic features with visual features. Instead, an interaction-based model should be used to directly encode the relationships between text lines.

Image of a Complex Table

project	budget	schedule
Redecoration of College of International Exchange	15,233,000	2022.12
Dormitory construction	56,000	2022.09

Extraction Result of the Table

project	budget	schedule
Redecoration of College of International	15,233,000	2022.12
Exchange		
Dormitory construction	56,000	2022.09

Ground Truth of the Table

project	budget	schedule
Redecoration of College of International Exchange	15,233,000	2022.12
Dormitory construction	56,000	2022.09

Figure 1. An image of a complex financial table with wrapped text (translated from Chinese) and the table extraction output of Tencent's TE model GrabTab, as well as the ground truth of the table's structure.

An image enhancement method is a series of technologies that process the collected images to make them more suitable for identification and analysis [20]. In this paper, we regard the problem of recognizing wrapped text as a text semantic matching task, for which an interaction-based language model is used to predict whether two text pairs belong to a wrapped text. We also propose ACTE, an image enhancement method that obtains and adds possible demarcation lines as an instruction for image-based table extraction methods to improve their performance on complex financial tables. Our contributions may be summarized as follows:

(1) We propose an image enhancement framework utilizing an OCR model, a language model, and heuristic algorithms to obtain row demarcation lines with high precision. This could serve as an instruction for end-to-end table extraction methods.
(2) We utilize an interaction-based model for predicting the semantic relationships between neighboring text pairs with higher accuracy to better recognize table cells with wrapped text.
(3) We develop a text semantic matching dataset for multi-line text recognition containing over 50,000 text pairs, as well as a complex financial table dataset containing over 600 tables.

The remainder of this article is structured as follows: Section 2 briefly summarizes related works on table structure recognition and text semantic matching. Section 3 explains the methodology of our method in detail. Section 4 presents a description of our dataset and experimental setup, along with experimental results and analysis. Finally, our conclusions are presented in Section 5.

2. Related Work

2.1. Table Structure Recognition

In the field of table structure recognition, early methods included heuristics and simple machine learning techniques. Itonori [1] proposed a method based on parallel line encircling and text block extension. This method involved the use of connected component bounding boxes to detect text boxes and segmentation lines. Text boxes were then extended to the segmentation lines and boundary line extensions of neighboring text boxes. Chandran and Kasturi [2] used continuous pixel points, instead of the commonly used Hough transform, to identify straight lines in table images. Rahgozar et al. [3] proposed a method based on gap clustering. Using this method, text blocks are first identified in the image; the gaps between them are then clustered to obtain the horizontal and vertical segmentation gaps in the form. Hassan and Baumgartner [4] proposed a recognition method based on table space layout templates. They classified tables into three types: tables with horizontal and vertical demarcation lines; tables with horizontal demarcation lines; and tables without demarcation lines. The rows and columns of the table were determined according to the position of the demarcation lines, the size of the text box, the length of the text, and other characteristics. Oro and Ruffolo [5] proposed a bottom-up heuristic method called PDF-TREX for recognizing table structures, using spatial coordinate clustering and the overlap relationship of text boxes in the horizontal direction. Kieninger et al. proposed T-Recs [6] and T-recs++ [7] to identify the structure of a table using a series of heuristic and unsupervised learning methods, including neighborhood unit clustering and column decomposition.

In the past decade, deep learning models have become the mainstream method for table structure recognition due to the booming development of visual deep learning models. Schreiber et al. [8] regarded table structure recognition as an image semantic segmentation task and proposed the DeepDeSRT system, which is based on an FCN (fully convolutional network) [9]. DeepTabStR [10] treats table structure recognition as a target detection task

and utilizes DCNs (deformable convolutional networks). The structure of a table is obtained by first extracting features using "deformable convolution" and "Rolpooling", then expanding the feature map to the size of the original image using "deconvolution" and "upsampling", and finally classifying each pixel. Prasad et al. [11] proposed CascadeTabNet, which employs CascadeRCNN to detect text boxes and then performs instance segmentation to determine the table's structure. Chi et al. [12] introduced a GNN (graph neural network) into the table structure recognition task and proposed the GraphTSR model. In this model, the table is considered as an undirected graph in which the text boxes are the nodes of the graph, and the visual features, such as the size and relative coordinates of the text boxes, are the attributes of the nodes. The structure of the table is recognized by predicting the relationships between the nodes. Li et al. [13] proposed RARE, an image description method based on the attentional mechanism. When an image is input into RARE, the model generates a string of HTML characters that define the structure of the table. Parsing this string of HTML characters provides the structural information on the form. TableMaster [14] is another attention-based method, which consists of two branches. One branch is used to predict the HTML sequence of the table structure; the other is used to predict the position and content of text boxes utilizing PSENet [15] and Master [16]. Li et al. [17] proposed UTTSR, which utilizes DBNet [18] to detect text lines, an attention-based CRNN (convolutional recurrent neural network) to identify text, and TableMaster to produce table structure sequences. Finally, LLMs (large language models) provide another possible means of table structure recognition [19]; using an LLM, tables are copied and pasted as model input; the desired information is then obtained by asking the model questions. However, the use of this method is limited to non-scanned PDF documents.

In recent years, some researchers have recognized that visual information alone may not be sufficient to accurately identify the structures of complex forms with wrapped text. Supplementary information needs to be introduced; specifically, textual semantic information. In light of this recognition, multimodal deep learning models have become a new trend in the field of table structure recognition. Zhang et al. [21] fused visual and textual information and combined bottom-up and top-down ideas in their proposed SEM model. The SEM model uses the ResNet and FCN models to divide the image of a table into grids. Then, the ROIAlign and BERT models embed the visual and semantic information, respectively, of the grids. Finally, the two feature vectors are combined, and the GRU (gated recurrent unit) model is used to determine the merging between the grid pairs. Li et al. [22] proposed the GFTE model, which is based on GraphTSR [12]. The model uses an LSTM model to semantically embed the text of each node; semantic vectors are then added to the attributes of the nodes, and a GCN (graph convolutional neural network) is used to predict the layout relationships between the nodes.

2.2. Text Semantic Matching

Text matching is a common task in natural language processing. Initially, statistical methods such as bag-of-words, TF-IDF, and BM25 [23] were used for text matching. However, these traditional methods exhibit certain shortcomings that cannot be ignored. Firstly, they do not consider contextual relationships among words, which may significantly impact the semantics of a sentence. For instance, the phrases "airplane from Seoul to Tokyo" and "airplane from Tokyo to Seoul" contain identical words, but their different orderings result in completely different meanings. Additionally, using similarity as a measure of text matching is not always appropriate. For example, in question-answering tasks, the relationship between text pairs is that of "question and answer" rather than "similarity". Finally, these conventional methods are inadequate for solving the problems of "polysemy" and "synonymy". For instance, the word "apple" can refer to either a fruit or a technology

company, while "China" and "sino" denote the same country. To address these issues, researchers have begun studying "text semantic matching", which involves matching text pairs based on their meaning rather than quantitative word characteristics. Depending on the model framework employed, text semantic matching models can be classified into two categories: representation-based models, and interaction-based models.

The representation-based model, also known as the dual-encoder model, utilizes semantic embedding models to produce the semantic vectors of text A and text B, respectively. It then calculates the degree of semantic matching between text A and text B based on the semantic vectors, using methods such as Euclidean distance, cosine similarity, or artificial neural networks. Microsoft's DSSR model [24] is a pioneering work in representational modeling. It studies the matching relationship between a query and document in the context of a web search. DSSR employs a bag-of-words model to embed the query and document. It then uses a DNN (deep neural network) to transform the sparse, high-dimensional embedding vectors into low-dimensional thick embedding vectors. Finally, it uses cosine similarity to compute the degree of matching between the two. More recently, several models have emerged which are consistent with DSSM in terms of framework, such as CDSSM [25], MV-LSTM [26], ARC-I [27], LSTM-RNN [28], and Sentence-BERT [29]. These models replace the semantic embedding or matching models with newer, more complex, and more effective ones. For instance, ARC-I utilizes CNNs (convolutional neural networks) to acquire semantic vectors and MLP (multilayer perceptron) to assess the degree of semantic matching. Sentence-BERT employs BERT to obtain semantic vectors and a Softmax classifier to predict matching degrees. The representation-based model has the advantages of supporting offline computation and fast matching speeds. However, its matching accuracy is often lower than that of the interaction-based model due to the loss of contextual information between texts caused by separate embedding and the limited semantic modeling ability of the matching layer.

The interaction-based model, also known as the cross-encoder model, abandons the idea of encoding and then matching that is used in the representation-based model. Instead, it allows text A and text B to interact directly, and then calculates the degree of matching between the two using interaction information. Inspired by image recognition, Pang et al. [30] proposed a CNN-based text matching model called MatchPyramid. The model embeds text A and text B as a list of word embedding vectors and calculates the interaction information between each position using three methods: Boolean, cosine similarity, and vector inner product. This produces an interaction matrix of size $m \times n$ (where m and n are the lengths of text A and text B, respectively). The matrix is then fed into a CNN to make predictions about their relationship. Vaswani et al. [31] proposed a transformer model that adopts a multi-head self-attention mechanism with position encoding. Transformer models can better capture the interaction information between different positions, and they effectively solve the long-range dependency problem associated with traditional models. Devlin et al. [32] proposed the BERT model, which is based on a bi-directional transformer structure with MLM (masked language model) and NSP (next sentence prediction) training methods. BERT can effectively capture contextual information, allowing for the words in each position of a text pair to fully interact with each other and thereby delivering a great improvement in semantic matching. Since then, following in the footsteps of BERT, numerous pre-trained language models based on transformer modules have been developed.

3. Methodology

In this paper, the ACTE (ante hoc enhancement for complex table extraction) method is proposed, consisting of two modules: a split module and a filter module. The overall framework of ACTE is shown in Figure 2. The split module takes an image of a table and

obtains its candidate row demarcation lines using an OCR (optical character recognition) model and a heuristic algorithm, dividing the table into grids. The filter module then uses a heuristic algorithm and a text semantic matching model to judge the correctness of every candidate demarcation line, one by one, from top to bottom. Next, false row demarcation lines are removed and true row demarcation lines are added to the input image. Finally, the processed image is input into downstream table extraction methods and the final extraction result is obtained.

3.1. The Split Module

The split module is based on an OCR model and a heuristic algorithm, as shown in Figure 3. The image of a table is regarded as an M × N 2-dimensional matrix, where M is the width of the image in pixels and N is the height of the image. A Cartesian coordinate system is established with the origin at the top-left vertex of the image, the *x*-axis along the top edge of the image, and the *y*-axis along the left edge of the image. First, PaddleOCR [33] is used to detect text boxes in the image, as shown in Equation (1). In this Equation, T is the set of all text boxes detected by the OCR model, t_i is a text box, c_i is the text line within the textbox, x_{i1} is the x-coordinate of the top-left vertex of the text box, y_{i1} is the y-coordinate of the top-left vertex of the text box, x_{i2} is the x-coordinate of the bottom-right vertex of the text box, and y_{i2} is the y-coordinate of the bottom-right vertex of the text box.

$$t_i = [c_i, x_{i1}, y_{i1}, x_{i2}, y_{i2}], t_i \in T \tag{1}$$

Second, the text boxes are projected onto the *x*-axis and *y*-axis, resulting in two arrays, p_x and p_y, indicating the number of text boxes on each coordinate for the *x*-axis and *y*-axis, respectively, as shown in Equations (2) and (3). The initial value of each p_i of the arrays is set as 0, and if a text box occupies a pixel corresponding to coordinate i, then the value of p_i is increased by 1.

$$p_x = [p_1, p_2, \ldots, p_M] \tag{2}$$

$$p_y = [p_1, p_2, \ldots, p_N] \tag{3}$$

Finally, candidate demarcation lines are obtained from the zero-value regions of p_x and p_y; thus, the image is split into grids. In detail, we extracted the continuous zero-value regions in p_x and p_y. For instance, if the p_x array of a certain image is [0, 0, 2, 2, 2, 0, 0], then its *x*-axis zero-value regions are [0, 1] and [5, 6]; this indicates that the pixels from x = 0 to x = 1 and from x = 5 to x = 6 are all blank, from which we can obtain the candidate column demarcation lines. If a zero-value region is on an edge of the image, the candidate demarcation line for the region is set as the edge line, such as x = 0 or x = M. If the zero-value region is not on the edge, then the candidate demarcation line is set as the middle line of the blank region corresponding to the zero-value region. For example, if the p_y array of a certain image is [0, 0, 1, 1, 0, 0, 0, 1, 1, 0, 0], then its zero-value regions are [[0, 1], [4, 6], [9, 10]] and the candidate row demarcation lines are {y = 0, y = 5, y = 10}.

3.2. The Semantic Matching Model

In this paper, we consider the recognition of wrapped text in complex tables as a unique text semantic matching task. The uniqueness of this task lies in the fact that the texts are often incomplete, and the relationship between the text pairs is based on semantic connectivity rather than semantic similarity, as is the case in traditional text-matching tasks.

Figure 4 shows our cross-encoder architecture, which uses a pre-trained language model to determine the semantic connectivity between text pairs. The model takes in a text pair as input and outputs a two-dimensional vector v, where v(0) represents the probability

of text a and text b being disconnected, and v(1) represents the probability of text a and text b belonging to the same wrapped text.

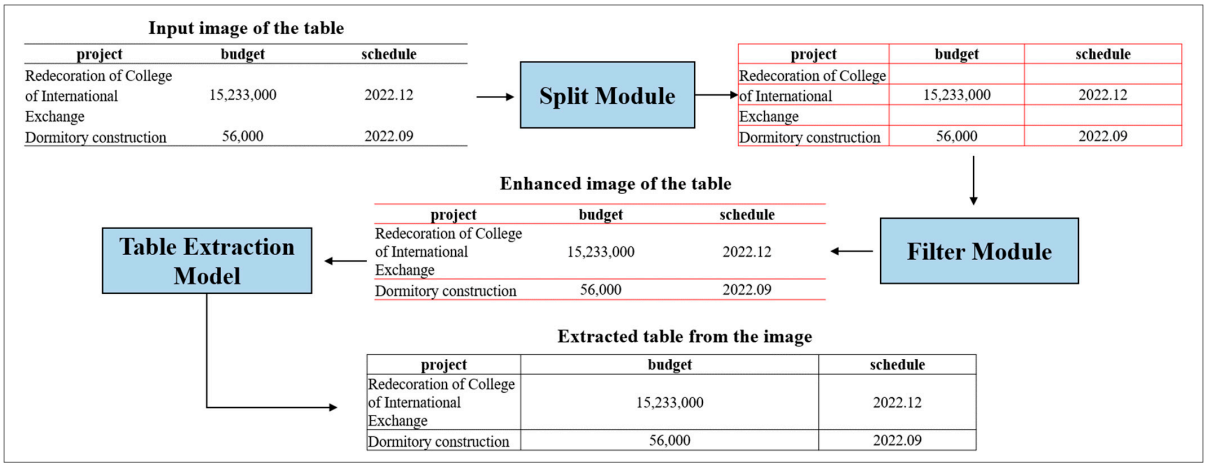


Figure 2. The overall structure of ACTE.

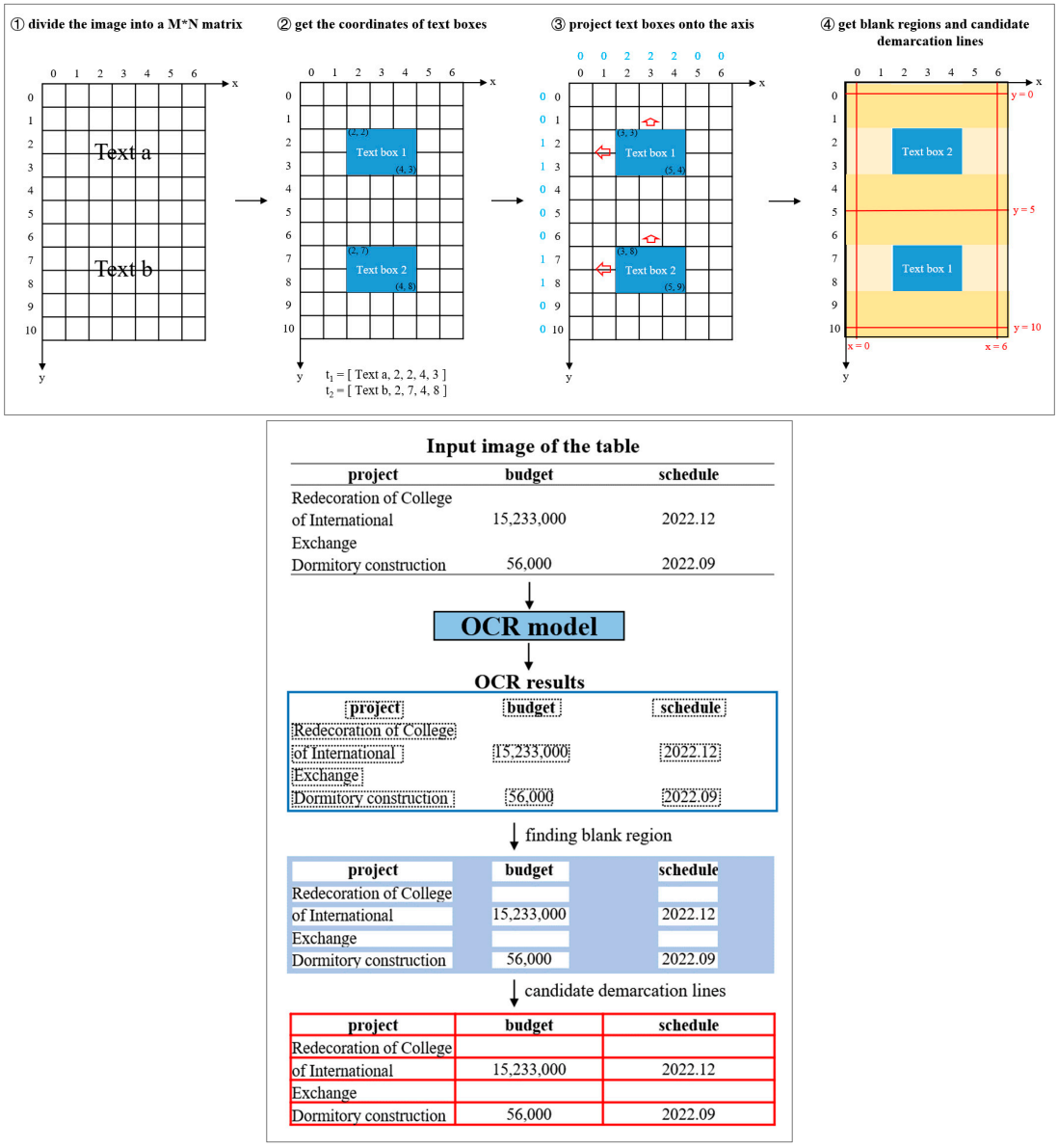


Figure 3. Processes of the split module.

3.3. The Filter Module

As column demarcation lines are usually irrelevant to textual semantic information and can be accurately detected using current TE methods, the split module does not filter them. However, they can be added to the image if required using the TE method.

After splitting the image into grids and completing the training of the semantic matching model, the filter module starts filtering the candidate row separators. We assume that whether a candidate row demarcation line is true can be judged according to whether there are neighboring text pairs above and below the line that belong to the same text. Thus, we initialize a sliding window with a height of two table rows defined by the top three candidate demarcation lines, x, y, and z, as shown in Figure 5. The sliding window then evaluates the correctness of line y and each candidate demarcation line is judged as it slides down from top to bottom, except for the top and bottom lines, which do not need to be disposed of, as shown in Algorithm 1. The top two rows within the windows are merged into one if line y is predicted to be false and remain unchanged if line y is predicted to be true. The use of a two-row sliding window turns the process into an iteration process which uses more information when predicting a demarcation line and helps to make the filter module work more precisely than when processing all candidate lines at one time. For instance, as shown in Figure 5, when the model handles the fifth candidate demarcation line, which is between text line 4 and text line 5, it makes decisions based on three pairs of texts: "Redecoration of College of International Exchange; Dormitory construction", "15,233,000; 56,000", and "2022.12; 2022.09". However, if the model is set to judge all candidate demarcation lines at one time instead of using the iteration process, it can only utilize one pair of incomplete texts: "Exchange; Dormitory", as the other two grids above the candidate demarcation line are blank, and this may lead to poorer performance.

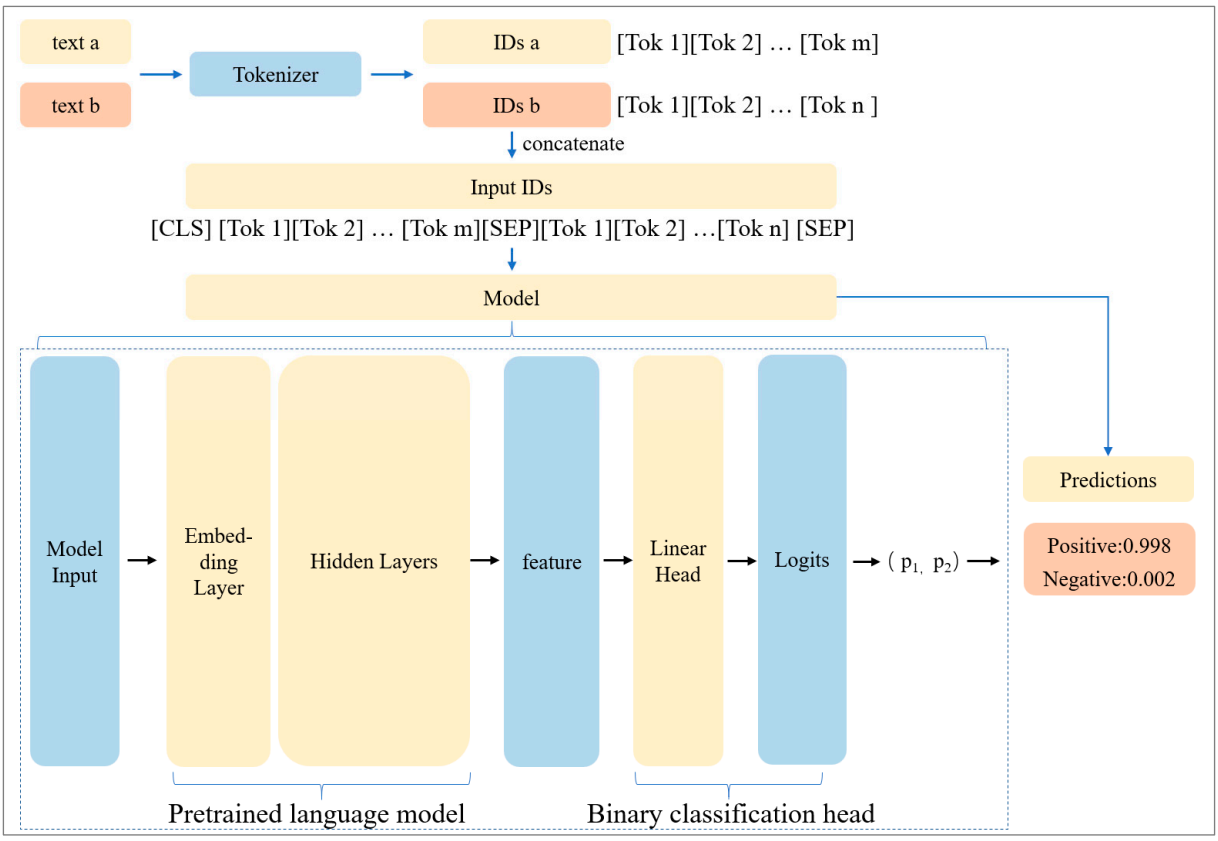


Figure 4. The architecture of the cross-encoder.

Algorithm 1. Filtering candidate row demarcation lines

Input:
textboxes: list of text boxes
candidate_lines: list of y-coordinate of candidate row demarcation lines
Output:
prediction: prediction of candidate row demarcation lines

```
x, y, z = 0, 1, 2
while z < len(candidate_lines) do
    line_x, line_y, line_z = candidate_lines[x], candidate_lines[y], candidate_lines[z]
    vertical_sep_lines = find_vertical_sep_lines(textboxes, line_x, line_y, line_z)
    number_of_districts = len(vertical_sep_lines) − 1
    f = [0] * number_of_districts
    for i in range(number_of_districts) do
        u_i, d_i = find_text_pairs(textboxes, i, Vertical_sep_lines, line_x, line_y, line_z)
        σ_i = semantic_matching_model(u_i, d_i)
        f[i] = σ_i
    end for
    if P(f) = 1 then
        prediction[y] = 1
        x = y
        y = z
        z +=1
    else
        prediction[y] = 0
        y = z
        z +=1
end while
return prediction
```

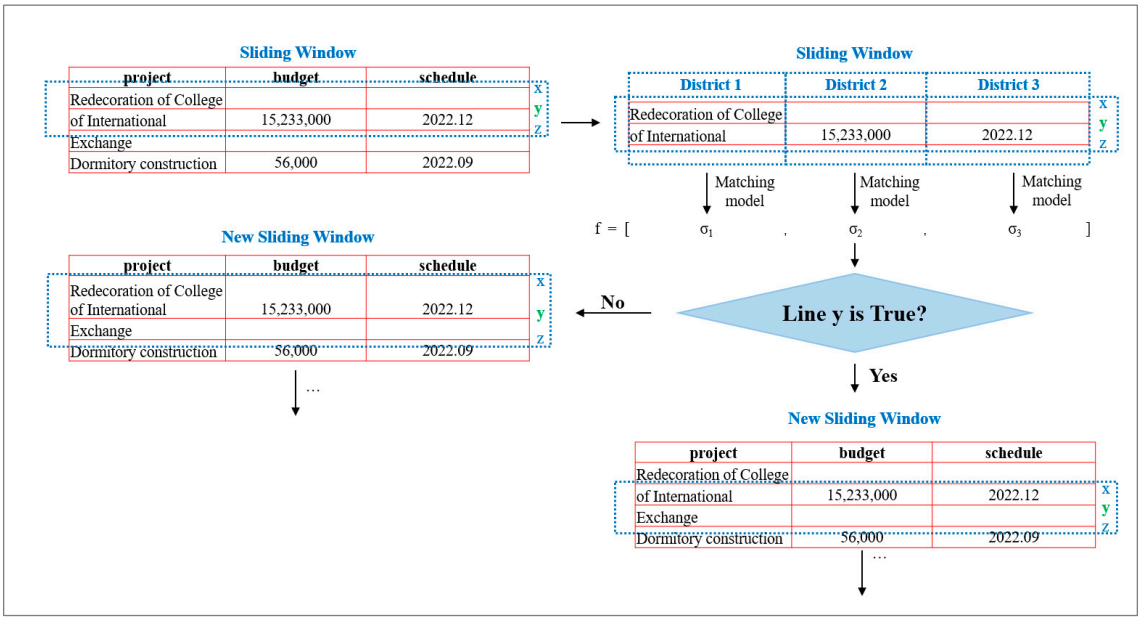


Figure 5. Processes of the filter module.

Specifically speaking, the sliding window divides the two table rows into several districts using column demarcation lines, as shown in Figure 5. Each district is further

divided using the split module to ensure there is only one table column per district. The trained semantic matching model is then utilized to determine the probability of text pairs within a district belonging to the same wrapped text. If either of the two grids is null, the probability is set to 0.5. After handling all districts, the correctness of the demarcation line y in the middle of the sliding window is determined by the maximum and minimum probabilities of all districts within the current sliding window. If the minimum probability is less than threshold a, demarcation line y is predicted to be true. If the maximum probability exceeds threshold b, demarcation line y is predicted to be false. Otherwise, demarcation line y is predicted to be true.

$$f_y = [\sigma_1, \sigma_2, \ldots, \sigma_t] \tag{4}$$

$$P(fy) = \begin{cases} 1 \ \min(f_y) < \text{thresholda} \\ 0 \ \max(f_y) > \text{thresholdb} \\ 1 \quad \text{otherwise} \end{cases} \tag{5}$$

If the candidate demarcation line y is predicted to be true, then x, y, and z will all move down one row to form the new sliding window. If line y is predicted to be false, the two table rows will be merged, and y and z will move down one row while x remains. Finally, after judging all candidate demarcation lines, the true row demarcation lines are added to the image of the complex table and the ante hoc process is completed.

4. Experiments

4.1. Dataset

While there are several public datasets available for table extraction tasks, they either lack complex tables, particularly those with wrapped text such as FintabNet [34], or they do not pertain to the financial field, such as SCI-TSR [12] and PubTabNet [35]. Because, in the present study, we concentrated on enhancing the extraction of complex financial tables, we created our own dataset by extracting images of tables from publicly available Chinese financial audit reports. This dataset comprised 671 images of complex tables, divided into two subsets: dataset A (564 images) and dataset B (107 images). Dataset A was utilized for training and evaluating semantic matching models, while dataset B was used to assess the effectiveness of ACTE for table extraction methods. An OCR model was employed to capture the text lines within the images of tables in dataset A. Neighboring text lines in the vertical aspect were turned into text pairs and labeled manually. The label was set to 1 if the two text lines belonged to the same wrapped text; otherwise, it was set to 0. The dataset contained over 16,000 pairs of adjacent text lines and was divided into a training set and a test set in a 7:3 ratio. Additionally, to prevent overfitting of the model, we expanded the training set to over 50,000 pairs using random text truncation and random negative sampling as shown in Figure 6.

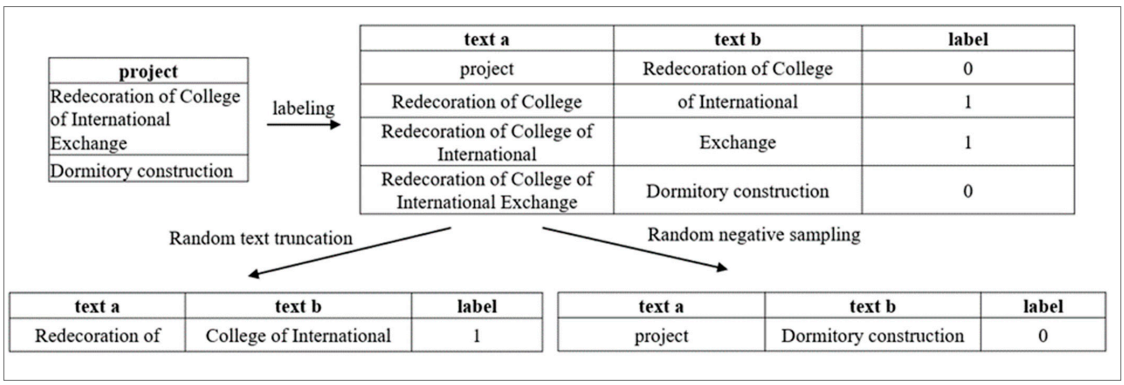


Figure 6. Processes of data labeling and data augmentation.

4.2. Semantic Matching Model

In the training and evaluation of semantic matching models, we conducted experiments using five state-of-the-art pre-trained language models: ERNIE-2.0 [36], ERNIE-3.0 [37], BERT-Chinese [32], M-BERT [38], and RoBERTa-Chinese [39]. Additionally, we developed two dual-encoder architectures, as shown in Figure 7, to demonstrate the advantage of interaction-based models over representation-based models in this task.

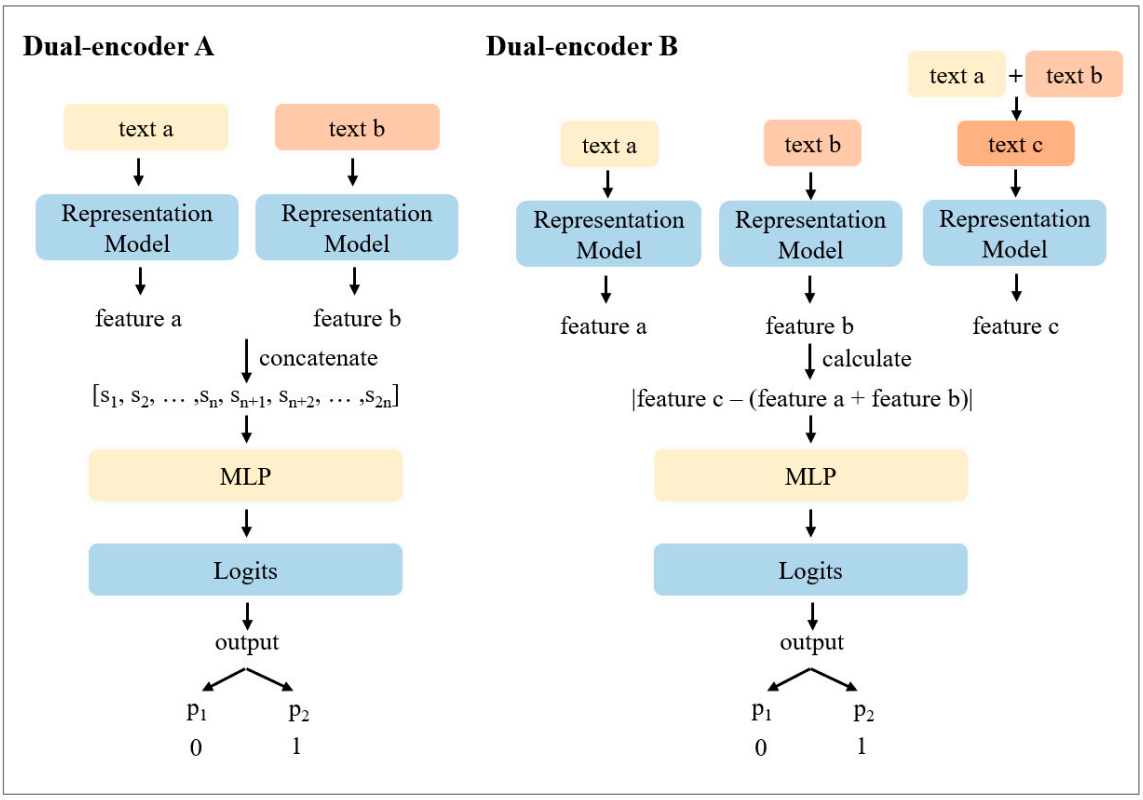


Figure 7. The architecture of the dual encoder.

Table 1 shows the AUC evaluation results for semantic matching models. The results indicated that cross-encoder models outperformed dual-encoder models in semantic connectivity matching. The best-performing model, based on ERNIE-3.0, was used to filter candidate demarcation lines in the following filter module.

Table 1. Results of the semantic matching experiment.

	Cross-Encoder	Dual-Encoder A	Dual-Encoder B
BERT	0.975	0.681	0.693
M-BERT	0.979	0.698	0.756
RoBERTa	0.973	0.657	0.635
ERNIE-2.0	0.981	0.703	0.714
ERNIE-3.0	0.983	0.705	0.723

4.3. Complex Financial Table Extraction Experiment

After obtaining the semantic matching model, we conducted a complex financial table extraction experiment on dataset B, which consisted of 107 images with 1127 table rows. Three state-of-the-art table extraction methods were employed in the experiment: GrabTab [40], PPstructureV2 [13], and Table Transformer [41]. GrabTab is a commercial table extraction method proposed by Tencent AI Lab in 2023 [40], which is currently the best-performing method according to the scientific complex table datasets SCI-TSR [12] and

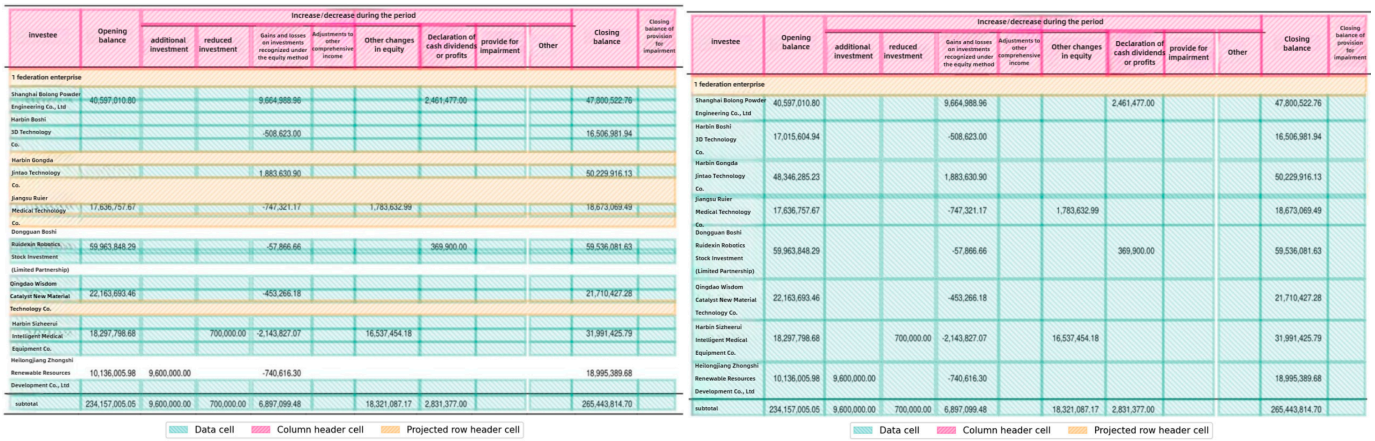
PubTabNet [35]. PPstructureV2 is an open-source method proposed by Baidu AI Lab in 2022 [13]; its performance currently ranks second on PubTabNet [35], after GrabTab. Table Transformer is an open-source table extraction method proposed by Microsoft in 2022 as a baseline for PubTables [41]. In the evaluation process, methods were manually assessed at the row level. A row in the table extraction result was labeled as a true positive if it matched the ground truth of the table, and labeled as a false positive otherwise. We used four evaluation metrics; these were recall score, precision score, F1 score, and TEDS (tree edit distance similarity), as shown in Equations (6)–(9), respectively. In the equations, T_a and T_b denote the table extraction result and its corresponding ground truth, respectively; EditDistance is the number of steps needed to transform T_a to T_b; and $|T|$ is the number of nodes in T. Examples are illustrated in Figure 8 and Supplementary Materials.

$$\text{recall} = \frac{\sum \text{true positive}}{\sum \text{true number of rows of the table}} \tag{6}$$

$$\text{precision} = \frac{\sum \text{true positive}}{\sum(\text{true positive} + \text{false positive})} \tag{7}$$

$$\text{F1} = \frac{2 \times \text{recall} \times \text{precision}}{(\text{recall} + \text{precision})} \tag{8}$$

$$\text{TEDS} = 1 - \frac{\text{EditDistance}(T_a, T_b)}{\max(|T_a|, |T_b|)} \tag{9}$$



(**a**) (**b**)

Figure 8. An example of the complex financial table extraction experiment. (**a**) The recognized table of the original image obtained using Table Transformer, in which only three rows are consistent with the ground truth of the table; (**b**) the recognized table of the preprocessed image obtained using Table Transformer, which is completely consistent with the ground truth of the table.

The experimental results in Table 2 show that our proposed ACTE method significantly improved the performance of these state-of-the-art table extraction methods. To be specific, ACTE increased the F1 score for ppstructure by 14.36, for Table Transformer by 11.39, and for GrabTab by 5.10. Theoretically, ACTE is applicable to all table extraction methods that take table demarcation lines as features, but not to methods that do not consider demarcation lines, such as GraphTSR [12] and GFTE [22].

To further evaluate ACTE's performance, we carried out a quantitative analysis of the extraction results obtained using GrabTab. We divided the condition of the extraction results into four kinds: successfully extracted rows with correct demarcation lines that were generated by ACTE; successfully extracted rows with incorrect demarcation lines;

unsuccessfully extracted rows with correct demarcation lines; and unsuccessfully extracted rows with incorrect demarcation lines. The percentage results for each condition are shown in Table 3.

Table 2. Results of the complex financial table extraction experiment.

	Recall	Precision	F1	TEDS
PPstructureV2	62.64	61.93	62.29	70.14
Table Transformer	82.96	79.57	81.23	84.47
GrabTab	90.77	90.29	90.53	89.63
PPstructureV2+ACTE	78.17	75.23	76.67	73.86
Table Transformer + ACTE	92.99	92.25	92.62	90.79
GrabTab+ACTE	96.18	95.09	95.63	92.53

Table 3. Percentage of the four kinds of condition in the extraction results.

	With Correct Demarcation Line	With Incorrect Demarcation Line
Successfully extracted	95.65%	0.53%
Unsuccessfully extracted	0.89%	2.93%

The results showed that 96.54% of the row demarcation lines generated by ACTE were correct and 98.6% of the extraction results were consistent with the row demarcation lines. This proved that ACTE performs well in obtaining row demarcation lines and that the lines on the processed image have a decisive impact on the extraction result of the downstream model; this explains why ACTE can enhance the performance of table extraction methods in extracting complex tables. These results also indicate that the precision of demarcation lines is a bottleneck that inhibits further improvements in the extraction results. This could explain why GrabTab saw the least improvement with ACTE—its base performance was so good that it quickly hit the bottleneck. Moreover, we found that over 50% of the incorrect demarcation lines appeared in table headers with multiple rows and spanning cells. How to improve ACTE's performance by strengthening its ability to handle table headers with complex structures is a question which needs to be explored in future studies.

4.4. Ablation Experiment

To evaluate the effect of the split and filter modules, we conducted an ablation experiment using GrabTab. The results, as shown in Table 4, indicated that the split module has a detrimental effect on the extraction of complex tables for GrabTab. Additionally, the filter module, which filters candidate row demarcation lines based on the semantic relationship between text lines, is essential in recognizing intricate table structures.

Table 4. Results of the ablation experiment.

	Recall	Precision	F1
GrabTab	90.77	90.29	90.53
GrabTab + SM	87.93	74.74	80.80
GrabTab + SM + FM	96.18	95.09	95.63

5. Conclusions

In this paper, we regard the problem of recognizing wrapped text in complex tables as a special semantic matching problem and propose the preprocessing method ACTE for complex financial table extraction. By utilizing an OCR model, an interaction-based

language model, and heuristic algorithms, ACTE makes full use of both visual information and textual semantic information to obtain the true row demarcation lines of a complex table and add them to the image. The effectiveness of this ante hoc enhancement method for current TE methods is demonstrated on a complex financial table dataset. The application of ACTE can save a lot of time for data scientists, financial professionals, and researchers engaged in the process of information extraction from financial documents.

One limitation of our research is that our dataset is relatively small and restricted to tables in financial statements due to the extremely high cost of manual data collection, processing, labeling, and evaluation. Nevertheless, the proposed framework could be applied to other documents and domains by fine-tuning the text semantic matching model using a corresponding corpus. Despite differences in contents and terminologies, tables in other documents exhibit structural patterns similar to those within our dataset; therefore, the overall framework and heuristic algorithms could still work.

Another limitation is that this method may not be effective for open-domain complex table extraction as the vocabulary increases and the semantic relationship between text pairs becomes more diverse and complex, making it harder to predict whether they belong to the same text.

Lastly, as shown in our experimental result, the precision of the demarcation lines generated by our method is the bottleneck for the final extraction results, with more than half of the incorrect predictions appearing in the headers of tables. Therefore, improving the ability of ACTE to deal with table headers is another important issue worthy of further study in the future.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/app15010370/s1. Figure S1. An example of the complex financial table extraction experiment. Rows that are inconsistent with the ground truth in the extraction result are highlighted in red. (a) An image of a complex financial table. (b) The ground truth of the financial table. (c) The extraction result for the original image obtained using GrabTab of the original image, which is labeled as 1 true positive and 2 false positives. (d) The enhanced image obtained using ACTE. (e) The enhanced image of the extraction result obtained using GrabTab, which is labeled as 4 true positives and 1 false positive.

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Data Availability Statement: Data available in a publicly accessible repository that does not issue DOIs. The Chinese complex financial dataset as well as the labeled dataset for training semantic matching models to recognize wrapped text in complex tables will be provided on https://github.com/yoiscarlet/ACTE.

Conflicts of Interest: The authors declare no conflicts of interest.

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